MD NAHIDUL ISLAM OPU

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EDUCATION

Doctor of Philosophy in Computer Science September 2026 - Present
The University of Manitoba, Canada

Master of Science in Computer Science September 2024 - August 2026
The University of Manitoba, Canada
CGPA - 4.25 / 4.50

Bachelor of Science in Computer Science & Engineering 2017 - 2022
Chittagong University of Engineering & Technology, Bangladesh (CUET)
CGPA - 3.45 / 4.00

EXPERIENCES

Grader May 2025 - August 2026
University of Manitoba *Winnipeg, Canada*

- Served as a grader for COMP-2450, COMP-1012, COMP-1010, COMP-2140
- Responsibilities were mainly grading projects, assignments, quizzes.

Sub-reviewer May 2025
35th IEEE International Conference on Collaborative Advances in Soft. and Comp. (CASCON 2025)

- Reviewed two technical papers.

Software Engineer November 2023 - July 2024
Tiger IT Bangladesh Ltd. *Dhaka, Bangladesh*

- Worked as a .NET developer.
- The main role was to develop features and maintain a thick desktop client.

Software Engineer July 2022 - October 2023
Enosis Solutions *Dhaka, Bangladesh*

- Worked as a full stack developer (.NET + React) and developed a crash reporting service website for Corsair. Designed the database, created REST APIs, and developed the website with React.

Junior Software Engineer (Part-time) July 2020 - October 2021
Arcade Studios *Dhaka, Bangladesh*

- As a game developer, Developed features of several games with Unity Engine and C# and integrated machine learning into a game.

SKILLS

Key Proficiencies	Software Engineering, Data Analysis, Experimental Design, Machine Learning and I
Programming Languages	C#, Python, Java, C++
Libraries & Frameworks	PyTorch, Hugging Face Transformers. scikit-learn, Flask, React, .Net, Unity3D

PUBLICATIONS

LLM-Based Detection of Tangled Code Changes for Higher-Quality Method-Level Bug Datasets
MSR'26 April 2026

- Formulated the problem as a binary classification task, evaluating prompting strategies (zero-shot, few-shot, chain-of-thought) using models like GPT-4o and Gemini-2.0-Flash.
- Achieved an F1-score of 0.88 with combined few-shot and chain-of-thought prompting; further improved to 0.906 using a multi-layer perceptron classifier on LLM-generated embeddings.

How Do Agentic AI Systems Address Performance Optimizations? A BERTopic-Based Analysis of Pull Requests
MSR'26 (Challenge Track) April 2026

- Identified 52 performance topics across 10 categories spanning the software stack using LLM classification and BERTopic on 1,221 AI agent-generated PRs.
- Performance PRs are rejected more often than non-performance ones, with UI/AI/Analytics faring worst and low-level best, and agents mostly optimize during feature development rather than maintenance.

How Do Agentic AI Systems Deal With Software Energy Concerns? A Pull Request-Based Study
MSR'26 (Challenge Track) April 2026

- The study analyzed energy-explicit AI agent-generated PRs via thematic analysis, deriving a 5-category taxonomy and 21 distinct optimization techniques, with most aligning with established energy research recommendations.
- Optimization PRs are accepted less often than other energy-aware PRs and have longer merge times, largely because they involve significantly larger, more complex changes that hurt maintainability.

Robotics Meets Software Engineering: A First Look at the Robotics Discussions on StackOverflow
Journal of Software: Evolution and Process April 2026

- analyzed robotics-related StackOverflow questions, finding that while volume is growing over time, robotics questions and answers are less popular than average—reflecting a small but engaged niche community.
- Thematic analysis revealed 11 major themes and showed that most of the questions are How questions, indicating practitioners mainly seek hands-on, step-by-step implementation guidance rather than conceptual understanding.

Understanding the Issue Types in Open Source Blockchain-based Software Projects with the Transformer-based BERTopic
CASCON'25 November 2025

- Applied topic modeling to understand the development trend in open-source blockchain based software projects with 1209 projects.
- Our findings suggest that blockchain field is still evolving with new problem and solutions which requires specialized tools, practices, and training for sustainable development.

Exploring Challenges in Test Mocking: Developer Questions and Insights from StackOverflow
Journal of Systems and Software December 2025

- Analyzed 25,302 StackOverflow questions and employed Latent Dirichlet Allocation for topic modeling, identifying 30 key topics grouped into five categories.
- Observed trends indicating shifts in developer priorities, with certain topics like Advanced Programming peaking between 2009 and 2012, while others like Mocking Techniques remained consistently dominant.

Moderately Mighty: To What Extent Can Internal Software Metrics Predict App Popularity at Launch?
arXiv preprint September 2025

- Created a dataset of 446 open-source Java-based Android apps consists of source code, code metrics, and user reviews.
- Applied regression and classification models to predict app popularity and concluded that internal software metrics are not enough but useful in predicting app popularity.

Handwritten Bangla Character Recognition using Convolutional Neural Networks: A Comparative Study and New Lightweight Model
Neural Computing and Applications — January 2024

- Analyzed performances of state-of-the-art CNN architectures in Bangla handwritten character recognition, proposed a new lightweight model and compared performances.

Towards quality improvement and prediction of closed questions on Stack Overflow
Algorithms for Intelligent Systems — October 2022

- Classified a question into one of the five classes according to the newly updated reasons after collecting the dataset from the Stack Overflow data dump and achieved state-of-the-art results focusing on feature extraction.

Learn2Write: Augmented Reality and Machine Learning-Based Mobile App to Learn Writing
MDPI Computers Journal — December 2021

- Integrated image classification to use machine learning for AR-based testing to detect an alphabet written on a board or paper and implemented on-screen handwriting teaching and testing features.

A Lightweight Deep Convolutional Neural Network Model for Real-Time Age and Gender Prediction
IEEE Xplore — December 2020

- A lightweight deep Convolution neural network model has been developed for real-time age and gender prediction.

PROJECTS

CUET BusKoi. A crowdsourcing-based Android application that collects passengers' locations and utilizes a clustering algorithm to track CUET Buses in real time and shows their locations using reverse geocoding. Built with React Native and Firebase real-time database. This app was made for the students to inform them of the current location of a bus.

Learn2Write. It is an Android app for kids that uses augmented reality and machine learning to teach handwriting in an innovative and fun way. Utilizing Unity, C#, and Python, image classification has been put into practice to evaluate handwritten characters written by kids on the screen.

BookBuddy. An Android app that allows users to exchange books. Users can enter book details they want to share and view books shared by others on this social site. Utilizing the comment section or direct messaging, users can communicate with one another. React Native and Firebase real-time database is used.

MediCare. An Android app, developed using Java and Android Studio, utilizes the Google Maps API and a PHP, MySQL backend. Its purpose is to assist users in finding doctors, hospitals, and ambulances through a map interface. Additionally, the app includes an emergency dial feature that allows users to contact the national emergency number or the nearest doctor.

EXTRA CURRICULAR ACTIVITIES

Mentor
CUET Computer Club

- Provided guidance to a group of students, offering expertise in diverse research and development areas
- Instructed them on initiating research as beginners and provided guidance throughout numerous project endeavors